

Lecture 23 -- Graphs and Distributions

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(Attendance Code: **837205**)

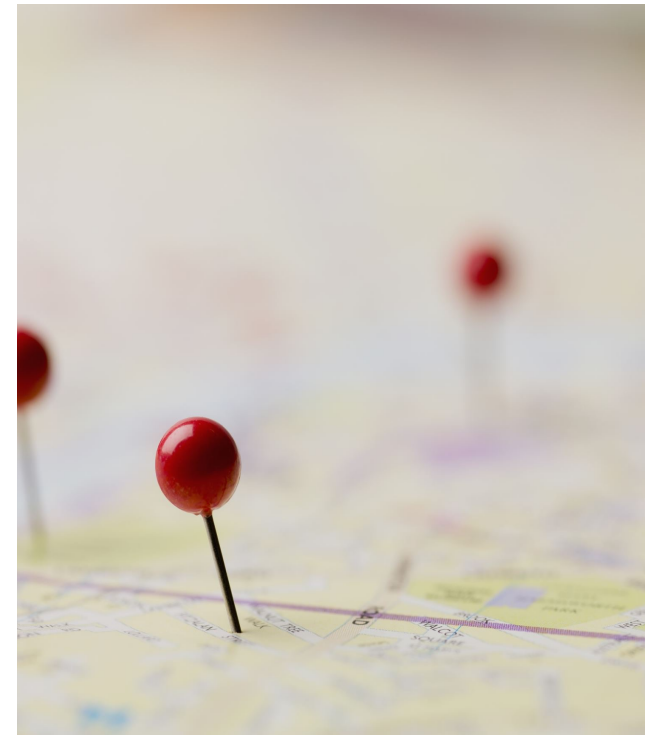
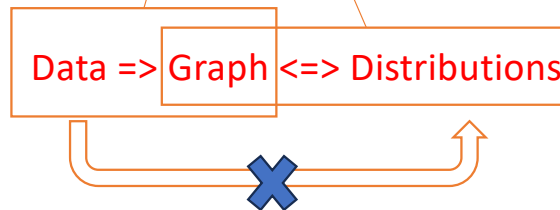
Up to now,

- Overview of Machine Learning
- Traditional Machine Learning Algorithms
- Adversarial Attack and Defence
- Deep learning
- Probabilistic Graphical Models
 - Introduction



Topics

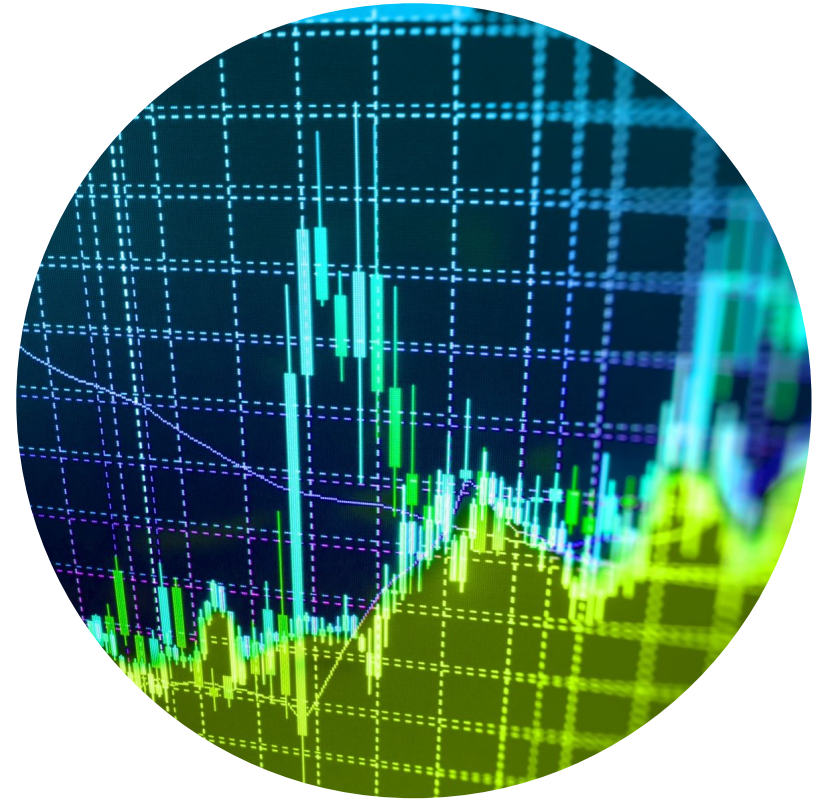
- Markov Assumption and Definition of I-Maps
- From I-Map to Factorization, and back
- Perfect Map
- Learning of Graph Models
- Caution in Establishing a Connection between Two Variables
- Overview of Structure Learning Methods



Graphs and Distributions

- Relating two concepts:
 - Conditional Independencies in distributions
 - Conditional Independencies in graphs
- I-Map is a relationship between the two

Graph $\Leftarrow$ I-Map $\Rightarrow$ Distributions



Recap: Conditional Independence

- X, Y independent $X \perp Y$ or $X \perp Y | \emptyset$

if and only if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

Now, given a joint distribution, we can determine the conditional independence according to the definitions.

- X and Y are conditionally independent given Z: $X \perp Y | Z$

if and only if:

$$\forall x, y, z : P(x, y | z) = P(x | z)P(y | z)$$

Independencies in a Distribution

- Let P be a distribution over X
- Define $I(P)$ to be the set of conditional independence assertions of the form $(X \perp Y | Z)$ that hold in P
- Example:

X	Y	P(X,Y)
x^0	y^0	0.08
x^0	y^1	0.32
x^1	y^0	0.12
x^1	y^1	0.48

X and Y are independent in P , e.g.,

$$P(x^1) = 0.48 + 0.12 = 0.6$$

$$P(y^1) = 0.32 + 0.48 = 0.8$$

$$P(x^1, y^1) = 0.48 = 0.6 \times 0.8$$

The same for $P(x^0, y^1)$, $P(x^0, y^0)$, and $P(x^1, y^0)$

Thus $(X \perp Y | \phi) \in I(P)$

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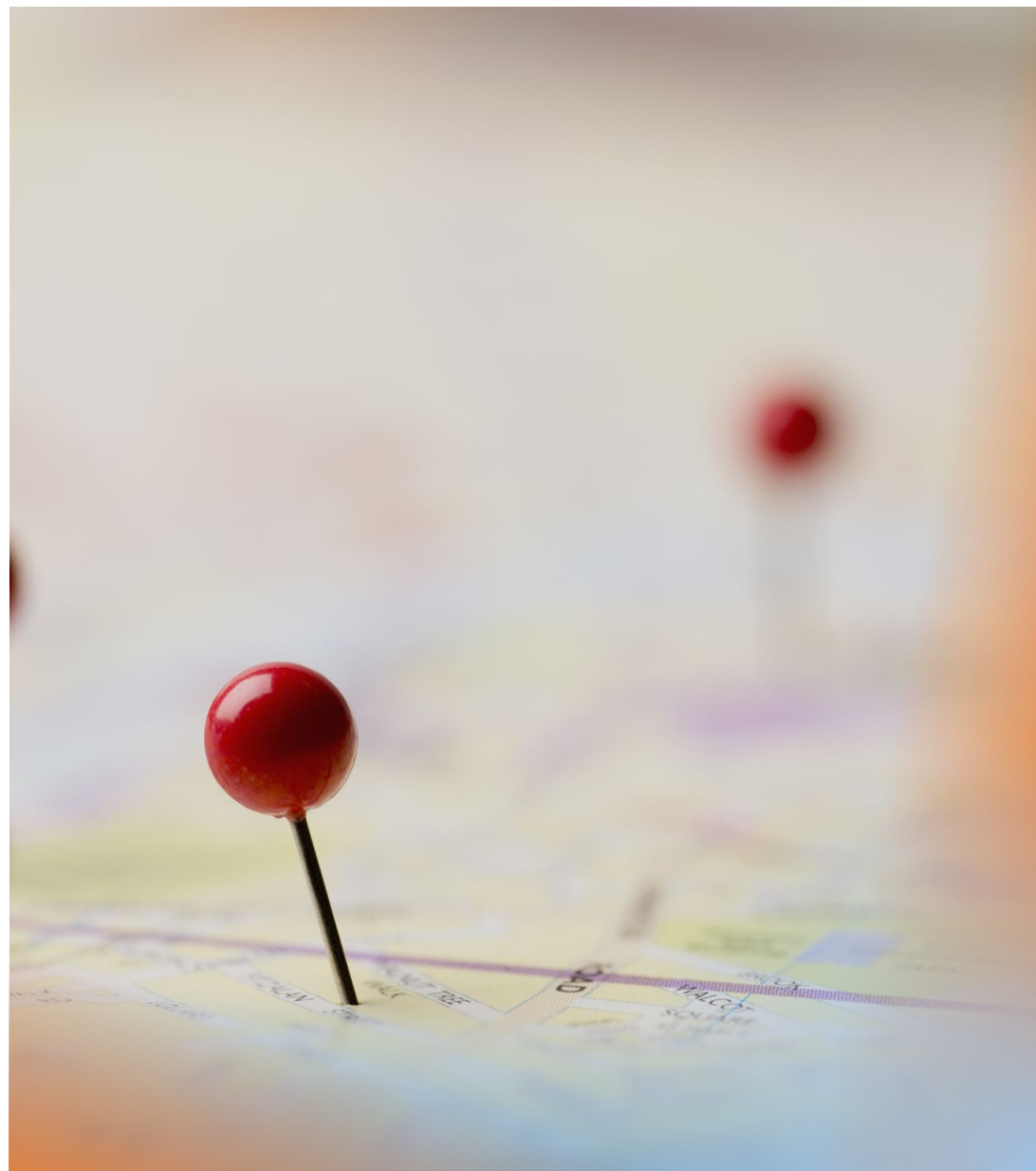
The same for $P(x^0, y^1)$, $P(x^0, y^0)$, and $P(x^1, y^0)$

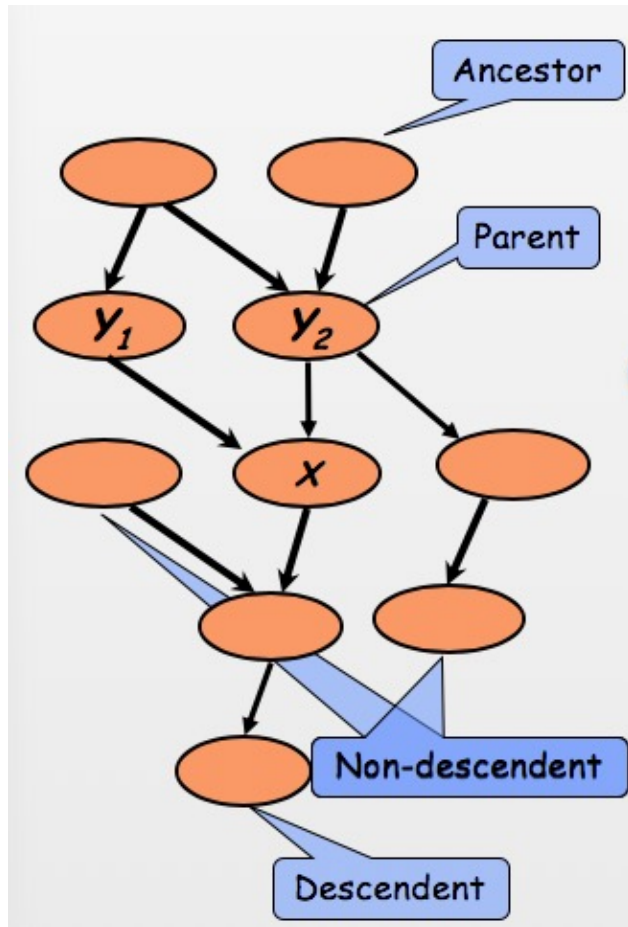
Thus $(X \perp Y | \phi) \in I(P)$

How about this distribution?

X	Y	P(X,Y)
x^0	y^0	0.10
x^0	y^1	0.16
x^1	y^0	0.64
x^1	y^1	0.10

Markov Assumption and Definition of I-Map



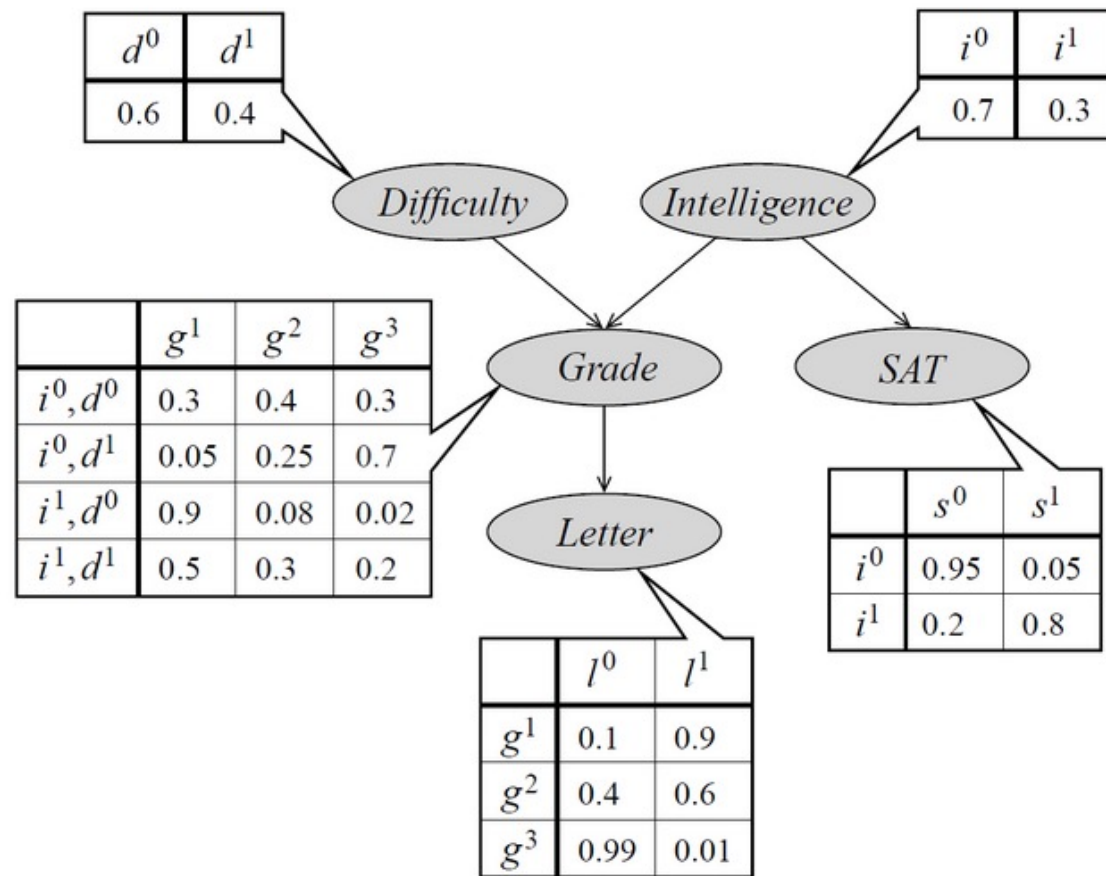


Markov Assumption

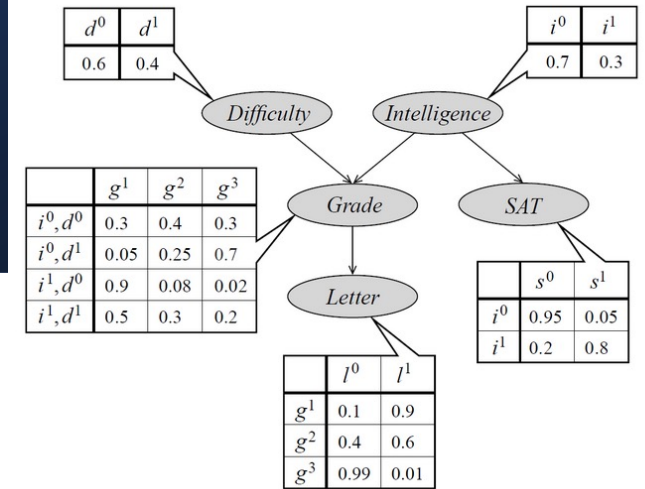
- We now make this independence assumption more precise for **directed acyclic graphs** (DAGs)
- Each random variable X , is independent of its non-descendants, given its parents $Pa(X)$
- Formally,

$$(X \perp NonDesc(X) | pa(X))$$

Can we read off
the
independencies
from a graph?



Independencies in a Graph



- Graph G with CPDs is equivalent to a set of independence assertions

$$P(D, I, G, S, L) = P(D)P(I)P(G|D, I)P(S|I)P(L|G)$$

- Local Conditional Independence Assertions (starting from leaf nodes):

$$I_{local}(G) = \{ (L \perp I, D, S | G), \quad L \text{ is conditionally independent of all other nodes given parent } G \\ (S \perp D, G, L | I), \quad S \text{ is conditionally independent of all other nodes given parent } I \\ (G \perp S | D, I), \quad \text{Even given parents, } G \text{ is NOT independent of descendant } L \\ (I \perp D | \emptyset), \quad \text{Nodes with no parents are marginally independent} \\ (D \perp I, S | \emptyset) \} \quad D \text{ is independent of non-descendants } I \text{ and } S$$

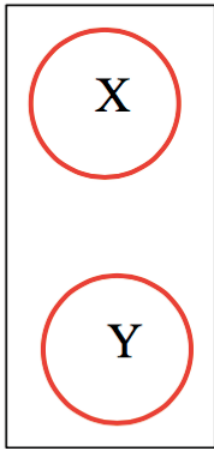
- Parents of a variable shield it from probabilistic influence
 - Once value of parents known, no influence of ancestors
- Information about descendants can change beliefs about a node

Definition of I-MAP

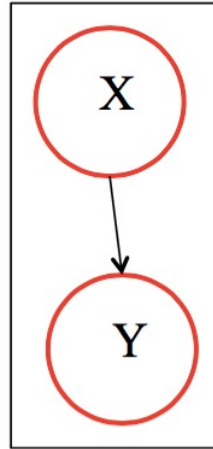
- Let G be a graph associated with a set of independencies $I(G)$
- Let P be a probability distribution with a set of independencies $I(P)$
- Then G is an **I-Map** of P if $I(G) \subseteq I(P)$
- From direction of inclusion
 - Distribution can have more independencies than the graph
 - Graph does not mislead in independencies existing in P
 - Any independence that G asserts must also hold in P
- $I_{\text{local}}(G)$, which includes all local Markov assumptions, is a subset of $I(G)$, which includes not only local Markov assumptions but also **global conditional independence**.
- $I_{\text{local}}(G) \subseteq I(G)$

to be introduced as d-separation

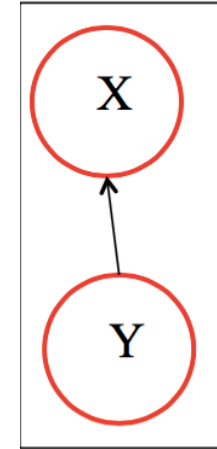
Example of I-MAP



G_0 encodes
 $X \perp Y$ or
 $I(G_0) = \{X \perp Y\}$

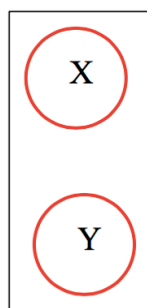


G_1 encodes no
Independence, or
 $I(G_1) = \emptyset$

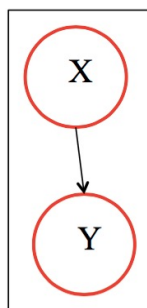


G_2 encodes no
Independence, or
 $I(G_2) = \emptyset$

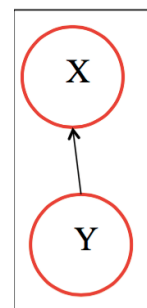
Example of I-MAP



G_0 encodes $X \perp Y$ or
 $I(G_0) = \{X \perp Y\}$



G_1 encodes no Independence, or
 $I(G_1) = \emptyset$



G_2 encodes no Independence, or
 $I(G_2) = \emptyset$

X	Y	$P(X,Y)$
x^0	y^0	0.08
x^0	y^1	0.32
x^1	y^0	0.12
x^1	y^1	0.48

X and Y are independent
in P , e.g.,

$$I(P) = \{X \perp Y\}$$

which suggests the following:

G_0 is an I-map of P

$$I(G_0) \subseteq I(P)$$

G_1 is an I-map of P

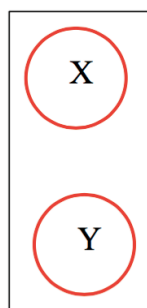
$$I(G_1) \subseteq I(P)$$

G_2 is an I-map of P

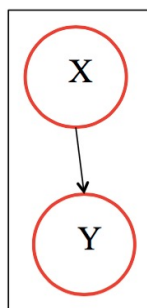
$$I(G_2) \subseteq I(P)$$

If G is an I-map of P then it captures **some** of the independences, not all

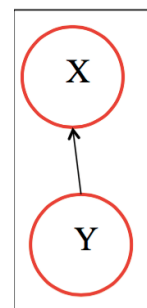
Example of I-MAP



G_0 encodes $X \perp Y$ or
 $I(G_0) = \{X \perp Y\}$



G_1 encodes no Independence, or
 $I(G_1) = \emptyset$



G_2 encodes no Independence, or
 $I(G_2) = \emptyset$

X	Y	$P(X,Y)$
x^0	y^0	0.4
x^0	y^1	0.3
x^1	y^0	0.2
x^1	y^1	0.1

X and Y are not independent in P

Thus $(X \perp Y) \notin I(P)$

G_0 is not an I-map of P

G_1 is an I-map of P

G_2 is an I-map of P

$I(P) = \emptyset$

which suggests the following:

$I(G_0) \not\subseteq I(P)$

$I(G_1) \subseteq I(P)$

$I(G_2) \subseteq I(P)$

If G is an I-map of P then it captures **some** of the independences, not all

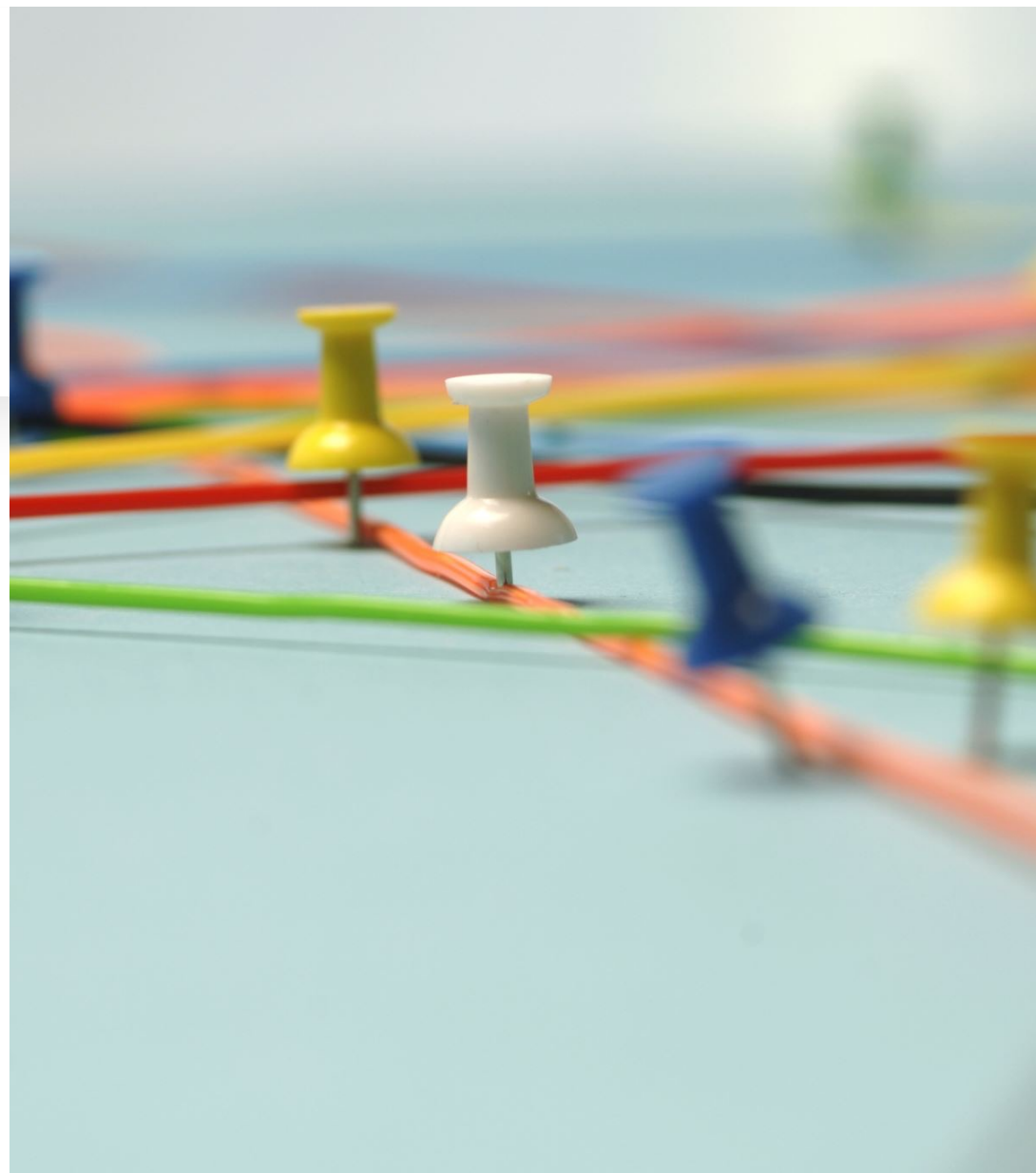
Exercise

- Please draw an I-Map for each of the following distributions:

x	y	P(x,y)
0	0	0.25
0	1	0.25
1	0	0.25
1	1	0.25

x	y	P(x,y)
0	0	0.2
0	1	0.3
1	0	0.4
1	1	0.1

From I-map to
Factorization, and back



What is factorization?

- **factorization** or **factoring** consists of writing a number or another mathematical object as a product of several *factors*, usually smaller or simpler objects of the same kind
- In our context, for example:

$$P(D, I, G, S, L) = P(D)P(I)P(G|D, I)P(S|I)P(L|G)$$

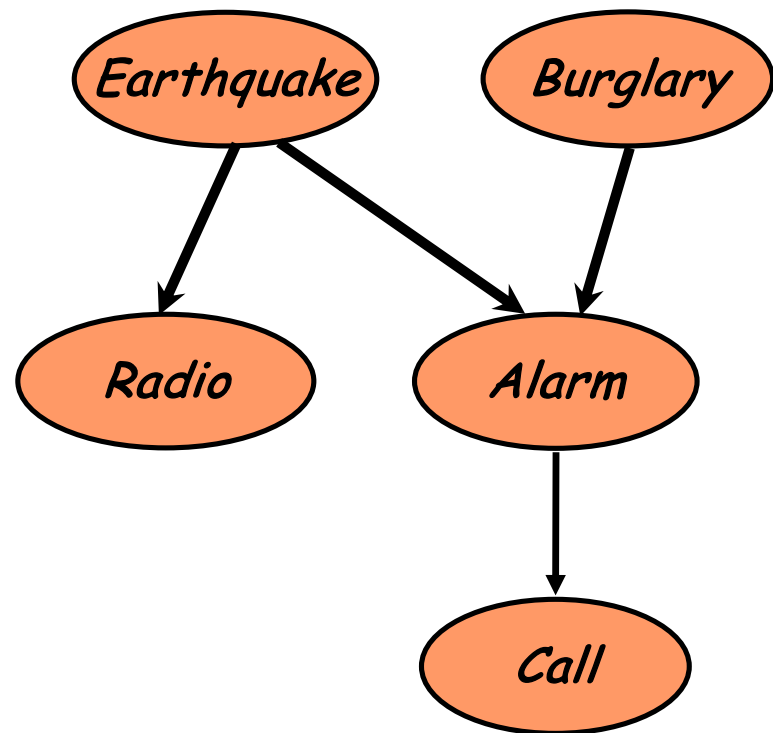
or

$$P(I, D, G, L, S) = P(I)P(D|I)P(G|I, D)P(L|I, D, G)P(S|I, D, G, L)$$

I-map to Factorization

- **Thm:** if G is an I-Map of P , then

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid Pa(X_i))$$



Exercise

- Please give the factorization of the distribution P according to the I-Map shown in the figure.

Factorization to I-map

Thm

$$P(X_1, \dots, X_n) = \prod_i P(X_i \mid Pa_i) \Rightarrow \mathbf{G} \text{ is an I-Map of } \mathcal{P}$$

Perfect Map



Perfect Map

- I-map
 - All independencies in $I(G)$ present in $I(P)$
 - Trivial case: all nodes interconnected
- D-Map
 - All independencies in $I(P)$ present in $I(G)$
 - Trivial case: all nodes disconnected
- Perfect map
 - Both an I-map and a D -map
 - Interestingly not all distributions P over a given set of variables can be represented as a perfect map



Learning of Graph Models

Two approaches to task of acquiring a model



1. Knowledge Engineering

Construct a network by hand with expert's help



2. Machine Learning

Learn model from a set of instances

Knowledge Engineering vs ML

Knowledge Engineering Approach

- Pick variables, pick structure, pick probabilities
- Too much Effort
 - Simple ones require hours of effort, complex one: months
- Significant testing of model by evaluating results of typical queries yield plausible answers

Machine Learning Approach

- Instances available from distribution we wish to model
- Easier to get large data sets rather than human expertise

Difficulties with Manual Construction

- In some domains:
 - Amount of knowledge required too large
 - No experts who have sufficient understanding
 - Cost: expert time is valuable
- Properties of distribution change from one site to another
- Change over time
 - Expert cannot redesign every few weeks
- Modeling mistakes have serious impact on quality of answers

Advantage of ML approach

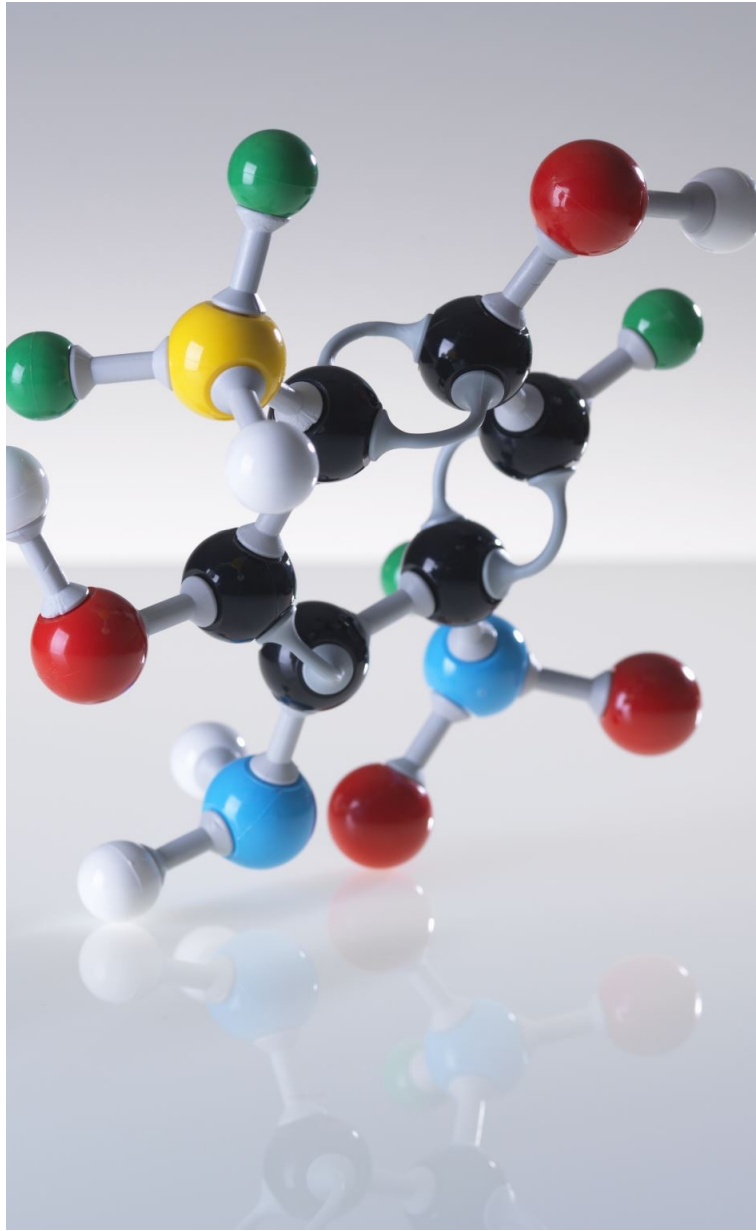
- We are in the Information Age
 - Easier to obtain even large amounts of data in electronic form than to obtain human expertise
- Example Data
 - Medical Diagnosis
 - Patient records
 - Pedigree Analysis (Genetic Inheritance)
 - Family trees for disease transmission
 - Image Segmentation
 - Set of images segmented by a person
- Machine Learning Algorithms

to work out the (conditional) probability tables



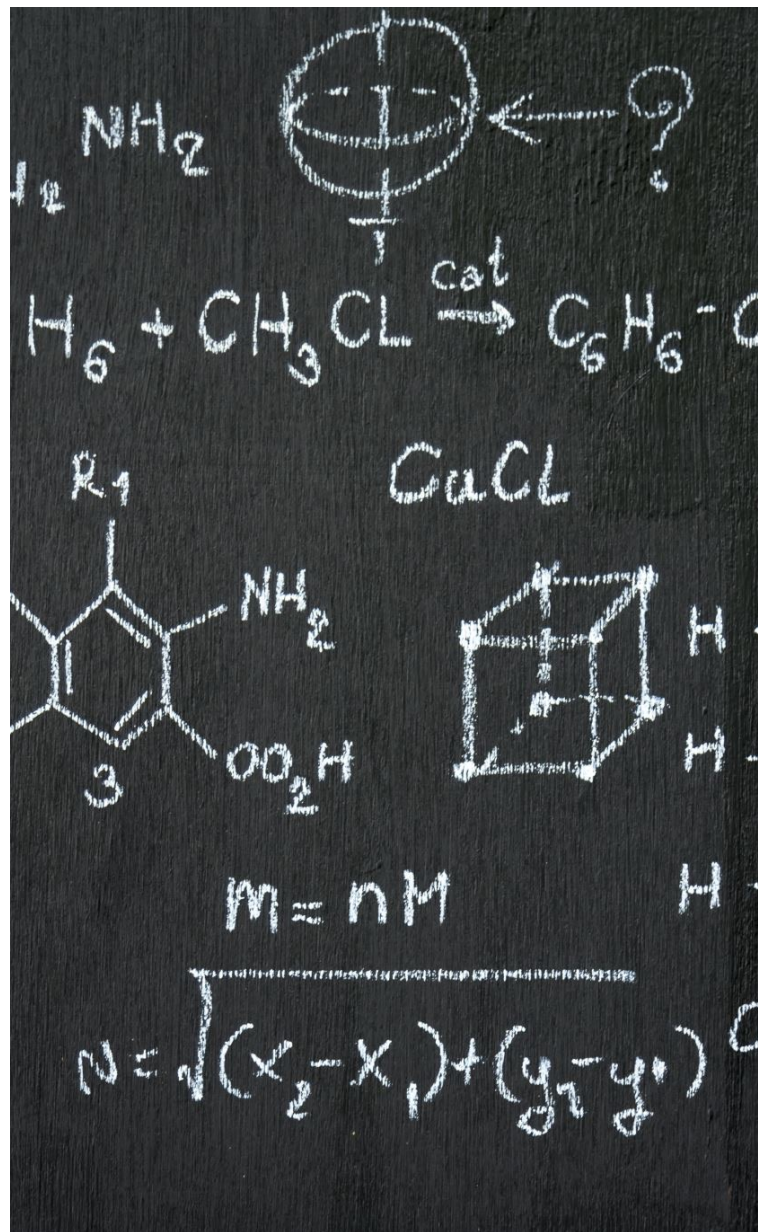
Example: Medical Diagnosis Task

- Collection of patient records
 - History:
 - Age, sex, history, medical complications
 - Symptoms
 - Results of tests
 - Diagnosis
 - Treatment
 - Outcome
- Task: **Use data to model distribution of patients**
 - Pathologist diagnoses disease of lymph nodes (Pathfinder 1992)



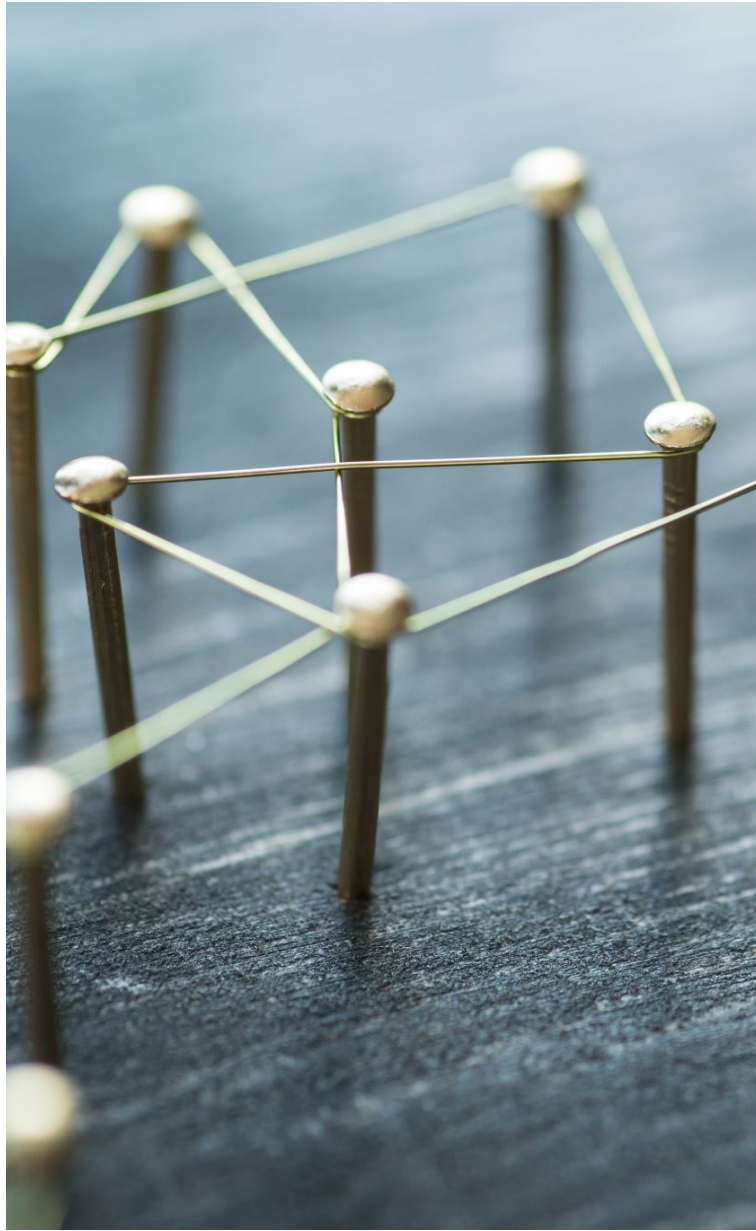
Goal of Structure Learning: Knowledge Discovery

- A tool for discovering knowledge about P^*
 - What are the direct/indirect independencies?
 - Nature of dependencies
 - E.g., positive or negative correlation
 - Example: in medical domain, which factors lead to a disease
- Bayesian network reveals much finer structure
 - Distinguish between direct and indirect independencies, both of which lead to correlations



Problem Assumptions

- We do not know the structure
- Dataset is fully observed
 - A strong assumption
- Assume data D is generated i.i.d. from distribution $P^*(X)$
- Assume that P^* is induced by BN G^*

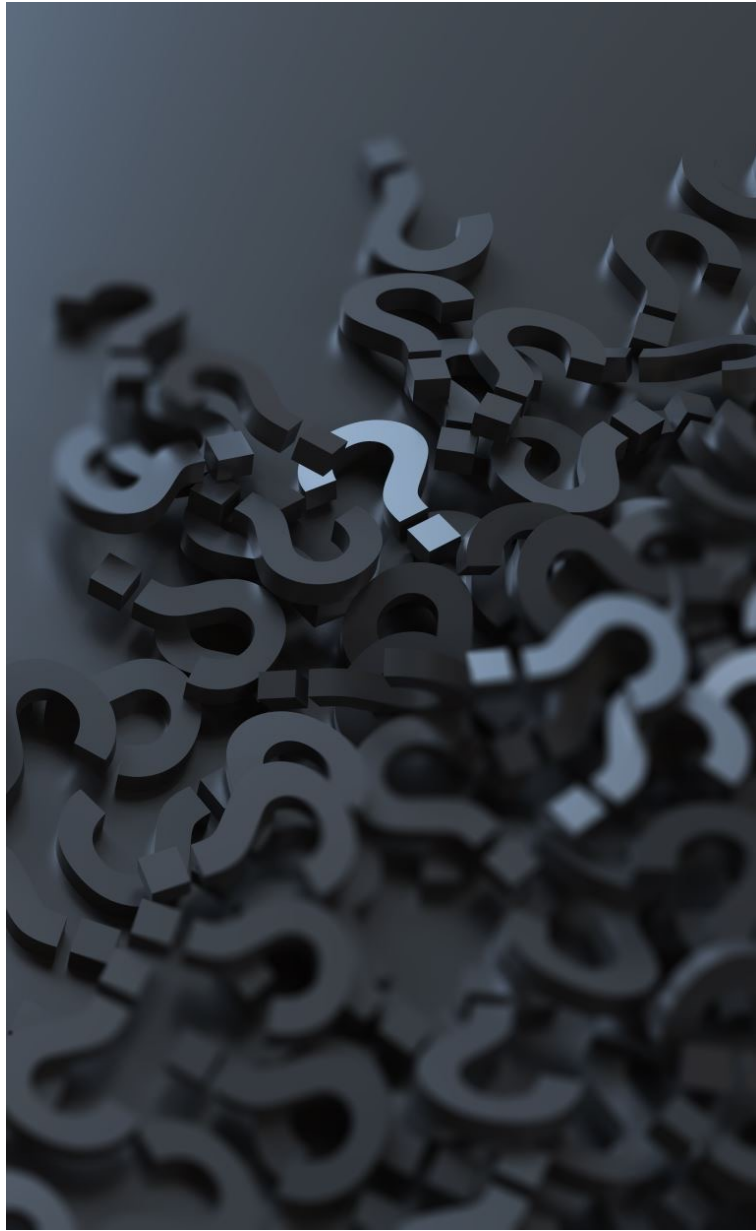


Caution in Establishing a Connection
between Two Variables



Knowledge Discovery Goal

- Goal: recover G^*
- Since there are many I-maps for P^* we cannot distinguish them from D
- Thus G^* is not *identifiable*
- Best we can do is recover G^* 's equivalence class



Too few or too many edges in G^*

- Even learning equivalence class of networks is hard
- Data sampled is noisy
- Need to make decisions about including edges we are less sure about
 - Too few edges means missing out on dependencies
 - Too many edges means spurious dependencies

To what extent do independencies in G^* manifest in D ?

- Two coins X and Y tossed independently
- We are given data set of 100 instances
- Learn a model for this scenario
- Typical data set:
 - 27 head/head
 - 22 head/tail
 - 25 tail/head
 - 26 tail/tail
- Are the coins independent?



Coin Tossing Probabilities

- Marginal Probabilities

- $P(X=\text{head})=.49$, $P(X=\text{tail})=0.51$, $P(Y=\text{head})=.52$, $P(Y=\text{tail})=.48$

- Products of marginals:

- $P(X=\text{head}) \times P(Y=\text{head}) = .49 \times .52 = .25$
- $P(X=\text{head}) \times P(Y=\text{tail}) = .49 \times .48 = .24$
- $P(X=\text{tail}) \times P(Y=\text{head}) = .51 \times .52 = .27$
- $P(X=\text{tail}) \times P(Y=\text{tail}) = .51 \times .48 = .24$

- Joint Probabilities

- $P(XY=\text{head-head})=.27$
- $P(XY=\text{head-tail})=.22$
- $P(XY=\text{tail-head})=.25$
- $P(XY=\text{tail-tail})=.26$

- But we suspect independence

- since probability of getting exactly 25 in each category is small (approx. 1 in 1,000)

27 head/head

22 head/tail

25 tail/head

26 tail/tail

According to
empirical
distribution: **not
independent**



Rain-Soccer Probabilities

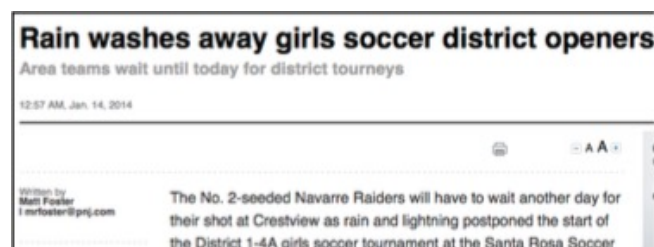
- Scan sports pages for 100 days
- Select an article at random to see
 - If there is mention of rain and soccer

- Marginal Probabilities

- $P(X=\text{rain})=.49$, $P(X=\text{no rain})=.51$, $P(Y=\text{soccer})=.48$, $P(Y=\text{no soccer})=.52$

- Joint Probabilities

- $P(XY=\text{rain-soccer})=.27$
 - $P(XY=\text{rain-no soccer})=.22$
 - $P(XY=\text{no rain-soccer})=.25$
 - $P(XY=\text{no rain-no soccer})=.26$



According to
empirical
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independent**



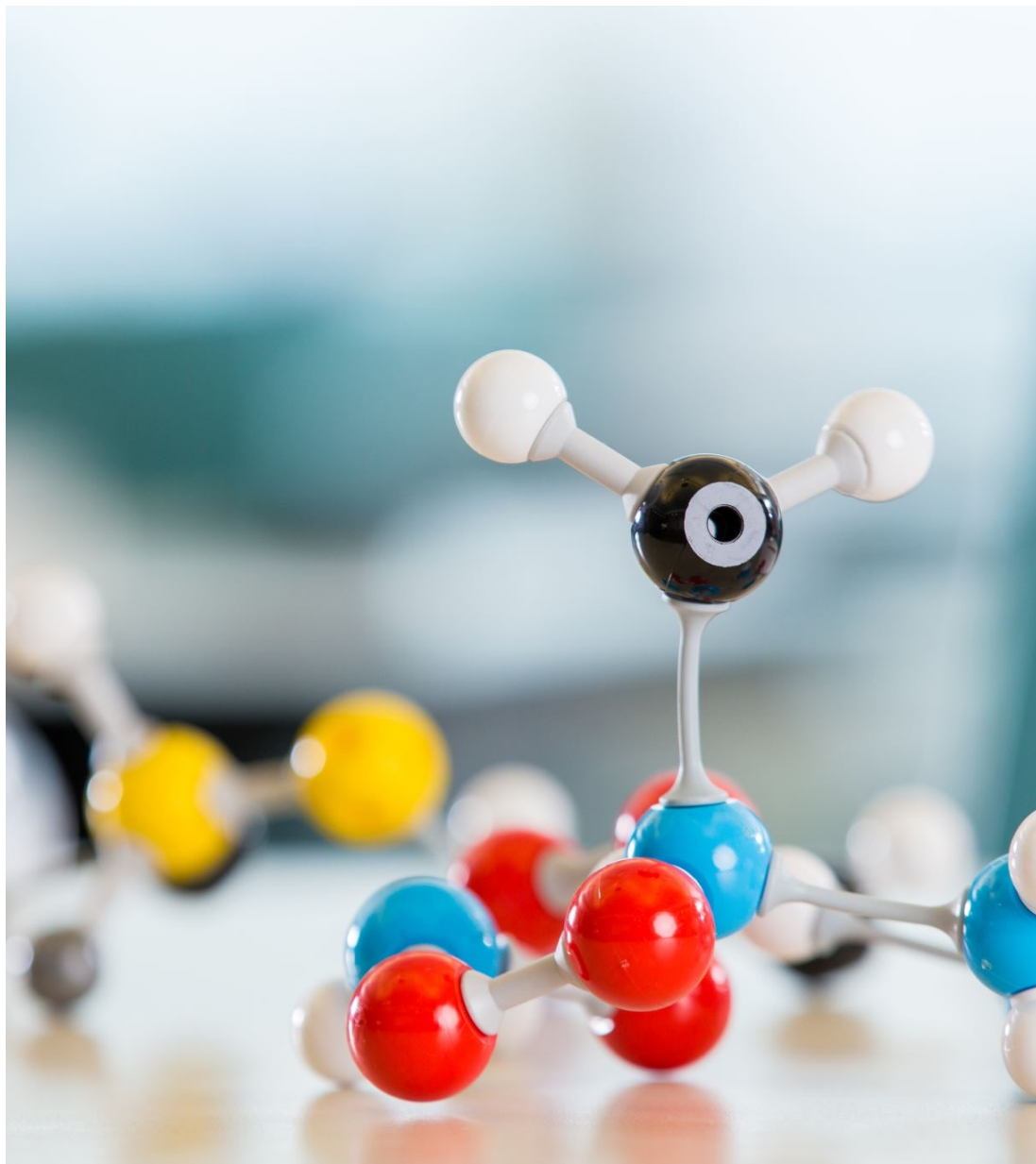
**We suspect there
is a weak
connection** (not
independent)

It is hard to be sure whether the true underlying model has an edge between X and Y

Data Fragmentation with spurious edges

- In a table CPD number of bins grows exponentially with number of parents
- Cost of adding a parent can be very large
- Cost of adding a parent grows with no of parents already there
- **It is better to obtain a sparser structure**
- We can sometimes learn a better model by learning a model with fewer edges even if it does not represent the true distribution.





Overview of Structure Learning methods

Structure Learning Algorithms

- Constraint-based

- Find structure that best explains determined dependencies
- Sensitive to errors in testing individual dependencies

Finds a Bayesian network structure whose implied **independence constraints** “match” those found in the data.

- Score-based

- Search the space of networks to find high-scoring structure
- Since space is super-exponential, need heuristics

Find the Bayesian network structure that can represent **distributions that “match”** the data (i.e. could have generated the data).

Elements of BN Structure Learning

Local: Independence Tests

- Measures of *Deviance*-from-independence between variables
- Rule for accepting/rejecting hypothesis of independence

Global: Structure Scoring

- Goodness of Network

Independence Tests

Idea: measure the independence between two variables with a value, and then determine whether to establish the connection based on the score.

- For variables x_i, x_j in data set D of M samples

- Pearson's Chi-squared (χ^2) statistic

$$d_{\chi^2}(\mathcal{D}) = \sum_{x_i, x_j} \frac{(M[x_i, x_j] - M \cdot \hat{P}(x_i) \cdot \hat{P}(x_j))^2}{M \cdot \hat{P}(x_i) \cdot \hat{P}(x_j)}$$

- Independence $d_{\chi}(D)=0$, larger value when Joint $M[x, y]$ and expected counts (under independence assumption) differ
- Mutual Information (K-L divergence) between joint and product of marginals

$$d_I(\mathcal{D}) = \frac{1}{M} \sum_{x_i, x_j} M[x_i, x_j] \log \frac{M[x_i, x_j]}{M[x_i]M[x_j]}$$

- Independence $d_I(D)=0$, otherwise a positive value

- Decision rule

- False rejection probability due to choice of t is its p-value

$$R_{d,t}(\mathcal{D}) = \begin{cases} \text{Accept } d(\mathcal{D}) \leq t \\ \text{Reject } d(\mathcal{D}) > t \end{cases}$$

Structure Scoring

Idea: measure each graph with a score, and select the graph with best score as G

- Log-likelihood Score for G with n variables

$$score_L(\mathcal{G} : \mathcal{D}) = \sum_{\mathcal{D}} \sum_{i=1}^n \log \hat{P}(x_i | pax_i) \quad \text{Sum over all data and variables } x_i$$

- 2. Bayesian Score

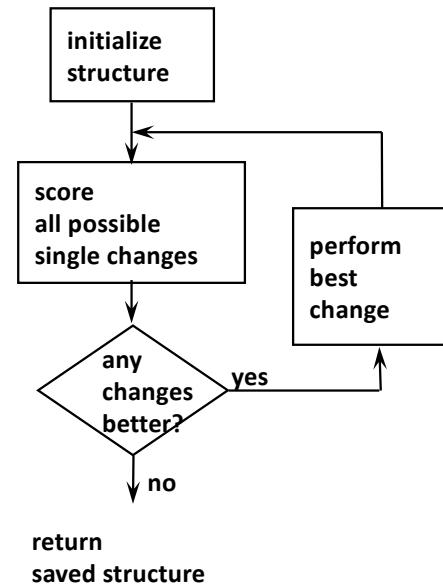
$$score_B(\mathcal{G} : \mathcal{D}) = \log p(\mathcal{D} | \mathcal{G}) + \log p(\mathcal{G})$$

- 3. Bayes Information Criterion

- With Dirichlet prior over graphs $score_{BIC}(\mathcal{G} : D) = l(\hat{\theta}_G : D) - \frac{\log M}{2} Dim(\mathcal{G})$

Model search

- Finding the BN structure with the highest score among those structures with at most k parents is NP hard for $k > 1$ (Chickering, 1995)
- Heuristic methods
 - Greedy
 - Greedy with restarts
 - MCMC methods



Heuristic for BN Structure Learning

- Consider pairs of variables ordered by χ^2 value
- Add next edge if score is increased

